

Pix2Pix for Nuclei Mask Generation from H&E-Stained Histopathology Images: Just the Baseline Approach

Azaliia Agisheva
Master in Artificial Intelligence
University of Vigo
Ourense, Spain

José Manuel Bueno Tabernero
Master in Artificial Intelligence
University of Vigo
Ourense, Spain

Abstract—We address the problem of automatically generating nuclei segmentation masks from stained histopathology images, framing it as a paired image-to-image translation task.

We implement a baseline Pix2Pix conditional Generative Adversarial Network (cGAN), composed of a U-Net generator with skip connections and a PatchGAN discriminator, and train it on the MoNuSeg dataset using the standard combined adversarial-plus-L1 loss.

The baseline model is trained for 28 epochs with early stopping and evaluated on a held-out test set using five image-quality metrics (SSIM, PSNR, MAE, RMSE, MSE). On the test partition the baseline obtains an SSIM of 0.709 and a PSNR of 10.67, providing a reference point against which subsequent architectural and loss-function improvements can be compared. We remark also that all code, trained weights, and a public inference demo are released on GitHub.

Index Terms—Generative adversarial networks, image-to-image translation, Pix2Pix, U-Net, PatchGAN, histopathology, nuclei segmentation.

I. INTRODUCTION

Automated nuclei segmentation in histopathology is a key enabling step for downstream tasks such as cancer grading, cell counting and computational pathology pipelines. Manual annotation is time-consuming and requires expert knowledge, which motivates data-driven approaches that learn to produce segmentation masks directly from raw tissue images [5].

This work tackles the problem as a *paired image-to-image translation*: given an stained tissue image, the model must generate the corresponding nuclei mask. We adopt the Pix2Pix conditional GAN framework of Isola *et al.* [3], which has shown strong performance on a wide range of paired translation tasks. The framework combines a generator that outputs the target image conditioned on the input and a discriminator that classifies whether an input-output pair is real or generated.

The contributions of this report (we only include the part regarding the Task 1 of the project due to report length limitations, the improvement task and results can be found on the notebook) are: (i) an implementation of the Pix2Pix baseline using a U-Net generator and a PatchGAN discriminator, adapted to the MoNuSeg dataset; (ii) a complete training and validation pipeline including reproducible data splits and early stopping; and (iii) a quantitative and qualitative evaluation on

a held-out test set, providing a reference performance level for the architecture, loss-function, data-augmentation and transfer-learning variants explored in subsequent tasks.

II. RELATED WORK

Generative Adversarial Networks (GANs) were introduced by Goodfellow *et al.* [1], who formulated image generation as a minimax game between a generator and a discriminator.

Conditional GANs (cGANs) [2] extend this idea by feeding both networks with an additional conditioning signal, enabling controlled generation. Pix2Pix [3] is the canonical cGAN for paired image-to-image translation: it uses a U-Net generator [4], which preserves spatial detail through skip connections, and a PatchGAN discriminator that classifies overlapping local patches rather than the whole image, encouraging high-frequency realism.

The training objective combines the standard adversarial loss with an L_1 reconstruction term, which has been shown to produce sharper outputs than L_2 [3], [9].

It is remarkable also that, on the histopathology domain, the MoNuSeg dataset [5] has become a standard benchmark for nuclear segmentation in multi-organ H&E images.

III. TECHNICAL APPROACH

A. Dataset and Preprocessing

We use the MoNuSeg dataset [5], which contains paired tissue/mask images. Each file stores the H&E input on the left half and the corresponding nuclei mask on the right half. We split each pair, resize both sides to 256×256 using bilinear interpolation, convert to tensors and normalise the pixel intensities to $[-1, 1]$ to match the tanh output of the generator.

The dataset is then split into 70% training (350 samples), 15% validation (75 samples) and 15% test (75 samples) with a fixed random seed for reproducibility.

B. Generator: U-Net

The generator follows the U-Net design of Ronneberger *et al.* [4]. The encoder consists of six downsampling blocks built on a 4×4 convolution with stride 2 (halving the spatial

resolution at each level), Batch Normalisation [6], and a LeakyReLU activation with slope 0.2 [7].

The first encoder block omits the normalisation, following the original Pix2Pix recipe. The channel progression is $3 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512 \rightarrow 512 \rightarrow 512$, reducing a 256×256 input to a 4×4 feature map.

A single 3×3 stride-1 convolution acts as the bottleneck, processing features at the lowest resolution without changing the spatial size.

The decoder mirrors the encoder using transposed convolutions with stride 2, Batch Normalisation and ReLU activations; the two deepest decoder blocks include Dropout(0.5), which both regularises the network and acts as the only source of stochasticity at inference, as recommended in [3].

After each decoder block, the output is concatenated with the corresponding encoder feature map (skip connection) before being passed to the next layer. A final 3×3 convolution maps the 64 decoder channels to 3 output channels, followed by a tanh activation that constrains the output to $[-1, 1]$. The generator has approximately 38.6 M trainable parameters.

C. Discriminator: PatchGAN

The discriminator is a PatchGAN [3], which classifies overlapping image patches as real or fake. Its input is the channel-wise concatenation of the conditioning image and the (real or generated) target, yielding a 6-channel tensor.

Four convolutional blocks (4×4 kernel, stride 2 for the first three and stride 1 for the fourth) progressively reduce the spatial resolution while increasing the channel count $6 \rightarrow 64 \rightarrow 128 \rightarrow 256 \rightarrow 512$; the inner blocks use Batch Normalisation and LeakyReLU. A final 4×4 stride-1 convolution produces a single-channel patch map, followed by a Sigmoid activation that yields per-patch real/fake probabilities. The discriminator has about 2.8 M parameters.

D. Loss Function

We use the standard Pix2Pix combined loss [3], with adversarial Binary Cross-Entropy (BCE) on the discriminator outputs and an L_1 reconstruction term on the generator outputs:

$$\mathcal{L}_G = \mathcal{L}_{\text{cGAN}}(G, D) + \lambda \mathcal{L}_{L_1}(G), \quad \lambda = 100, \quad (1)$$

$$\mathcal{L}_D = \frac{1}{2} (\text{BCE}(D(x, y), 1) + \text{BCE}(D(x, G(x)), 0)). \quad (2)$$

The large λ pushes the generator towards pixel-accurate outputs, while the adversarial term encourages local realism. As argued in [3], [9], L_1 is preferred over L_2 because it produces sharper images.

E. Training Procedure

The generator and discriminator are trained with the Adam optimiser [8] ($\beta_1 = 0.5$, $\beta_2 = 0.999$) at a learning rate of 2×10^{-4} . Each mini-batch (size 8) triggers one discriminator and one generator update in that order, with the discriminator’s input detached from the generator’s computation graph to ensure clean gradients. Validation loss is computed at the end of every epoch, and early stopping with a patience of 10

TABLE I
TEST SET PERFORMANCE METRICS FOR THE BASELINE PIX2PIX MODEL ON MoNuSeg (75 PAIRED SAMPLES).

Metric	Mean	Std. dev.
SSIM ($\uparrow$)	0.7089	0.0771
PSNR (dB, $\uparrow$)	10.668	1.481
MAE ($\downarrow$)	0.1072	0.0323
RMSE ($\downarrow$)	0.2971	0.0500
MSE ($\downarrow$)	0.0907	0.0304

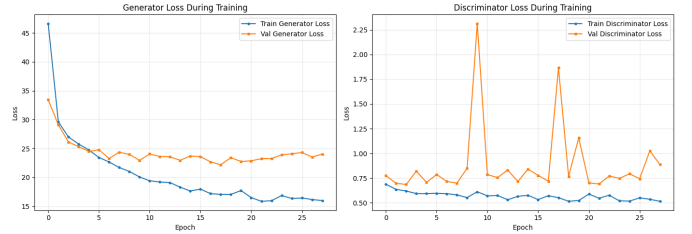


Fig. 1. Generator and discriminator training/validation loss curves for the baseline Pix2Pix model. Training converged after 28 epochs (early stopping with patience = 10).

epochs is used to avoid overfitting; in our experiments training converged after 28 epochs.

F. Public Repository

A public GitHub repository with the corresponding notebook, the full inference demo, the trained generator weights, and a curated subset of MoNuSeg samples is available at https://github.com/jmbuenot/CVII-Azaliia_JBT-Demo-Generator-Project.

IV. RESULTS

We evaluated the trained baseline on the 75 test samples using five image-quality metrics: the Structural Similarity Index (SSIM) [10], Peak Signal-to-Noise Ratio (PSNR), Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and Mean Squared Error (MSE). The aggregated results are reported in Table I. Notice that SSIM and PSNR are reported because they correlate better with perceived image quality, whereas MAE/RMSE/MSE quantify pure pixel-level fidelity.

The baseline achieves an SSIM of 0.709 ± 0.077 and a PSNR of 10.67 ± 1.48 on the test set. The per-sample distributions in Fig. 2 show that the model behaves consistently across the dataset, with SSIM values concentrated around the mean and a small left tail (worst case 0.540, best case 0.864). PSNR exhibits a wider relative spread, suggesting that some samples contain large local errors that an SSIM-based metric does not penalise as strongly.

The training dynamics are shown in Fig. 1. The generator loss decreases steadily during the first epochs and stabilises around epoch 15, while the discriminator loss settles at an equilibrium that indicates neither network has collapsed, consistent with what is reported in the original Pix2Pix work [3].

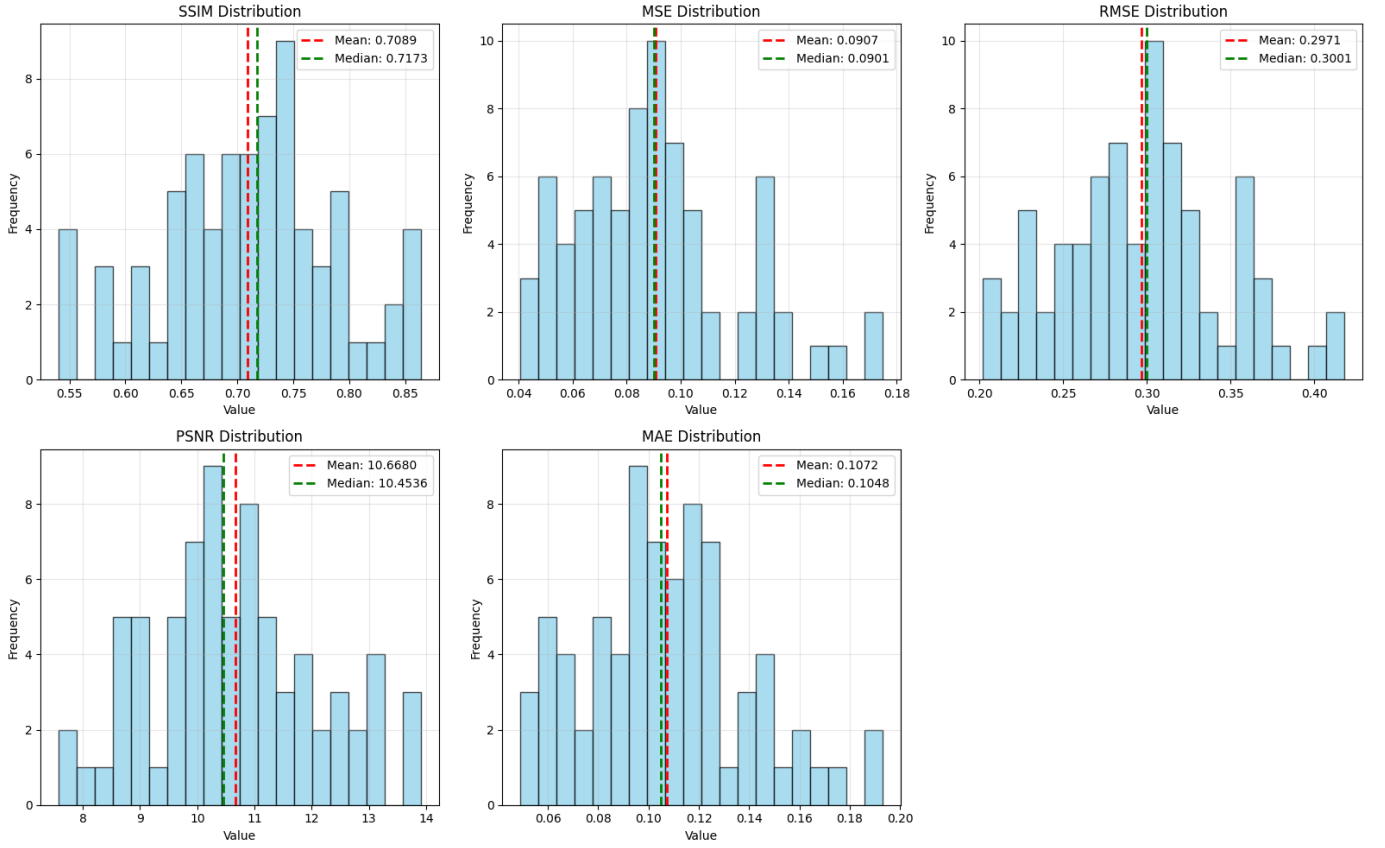


Fig. 2. Distribution of per-sample evaluation metrics on the test set (SSIM, MSE, RMSE, PSNR, MAE), shown with mean (red) and median (green) reference lines.

The early-stopping criterion ended training at epoch 28, when the validation generator loss stopped improving for ten consecutive epochs.

We can see (Fig. 3) how the model successfully captures the global distribution of nuclei in each tissue image. The generated masks reproduce the high-density regions correctly and follow the spatial layout of the ground truth. Nevertheless, some fine boundaries of small or overlapping nuclei are not perfectly delineated, which is reflected in the moderate per-sample SSIM values and in the relatively modest PSNR. These limitations serve as a motivation for the architectural, loss-function, augmentation and transfer-learning improvements explored in subsequent tasks of the project, which can be found on the notebook.

V. CONCLUSION

We have implemented and evaluated a baseline Pix2Pix model for the task of generating nuclei segmentation masks from stained histopathology images on the MoNuSeg dataset. The model combines a U-Net generator with a PatchGAN discriminator and is trained with the standard $\mathcal{L}_{cGAN} + 100 \mathcal{L}_{L_1}$ loss.

The baseline attains $\text{SSIM} = 0.709$ and $\text{PSNR} = 10.67$ on a held-out test set of 75 paired samples, providing a reproducible reference performance level. Qualitative inspection

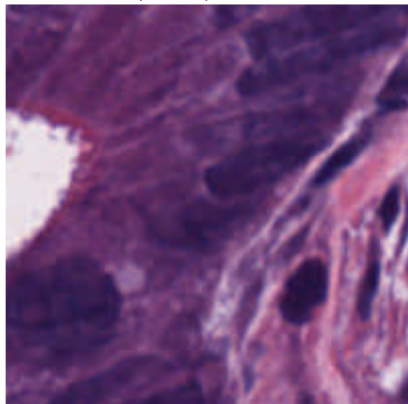
shows that the model captures the spatial distribution of nuclei reliably but still struggles with fine boundary detail in dense regions.

These observations motivate the next tasks of the project, in which we explore alternative generator architectures (ResNet, Attention U-Net), alternative loss formulations (LSGAN, L_2 , L_1 -only), paired data augmentation, and transfer learning from ImageNet pre-trained encoders. As already stated, the code, trained weights, and a small inference demo are publicly available at the URL given in Section III, Subsection F.

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Sample 1: Input Condition



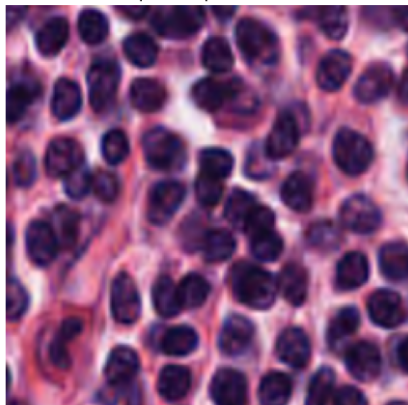
Generated Image



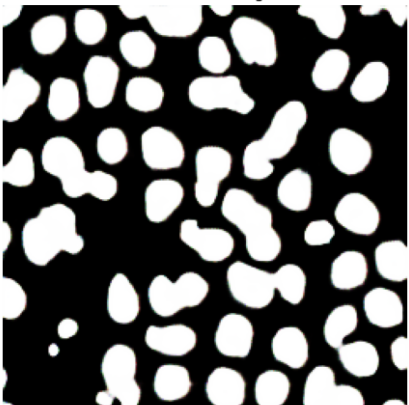
Real Image
SSIM: 0.7478
PSNR: 10.59
MAE: 0.0991



Sample 2: Input Condition



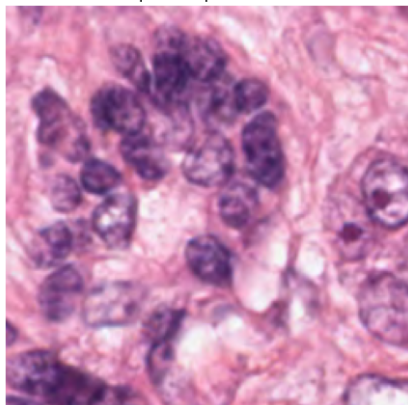
Generated Image



Real Image
SSIM: 0.5989
PSNR: 9.99
MAE: 0.1274



Sample 3: Input Condition



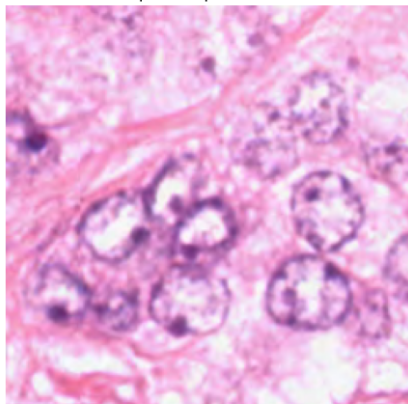
Generated Image



Real Image
SSIM: 0.7173
PSNR: 9.78
MAE: 0.1186



Sample 4: Input Condition



Generated Image



Real Image
SSIM: 0.6980
PSNR: 8.89
MAE: 0.1408



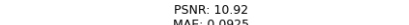
Sample 5: Input Condition



Generated Image



Real Image
SSIM: 0.7572
PSNR: 10.92
MAE: 0.0925



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